Retrofitting existing fossil-fuel heating systems in lightweight commercial buil

Comprehensive Hardware Design Analysis

Revision 1 | Generated 04/05/2026

Executive Summary

Generated

Project ID	**a_compact_commercial_airsource_heat_pump**
Industry Domain	**commercial_hvac**
Engine Version	**PDF Engine v2.1 (Full Detail)**
Target Quantity	**Not specified**

ARCHITECTURE AT A GLANCE

MODULES
0

BILL OF MATERIALS ROWS
0

IDENTIFIED RISKS
0

ECONOMICS

ESTIMATED UNIT COST
£0.00

TARGET CEILING
£4,500.00

HEADROOM
£4,500.00

TOTAL NON-RECURRING ENGINEERING
£0.00

UNIT COST PLUS AMORTISED NON-RECURRING ENGINEERING
£0.00

Decision Required

BLOCKED — Brief Incomplete

Required field missing: product_type; Required field missing: target_cost; Required field missing: production_volume; Required field missing: max_mass; Required field missing: jurisdiction; Required field missing: thermal_capacity_kw; Required field missing: cop_target; Required field missing: refrigerant_type; Required field missing: acoustic_target_dba; Required field missing: architecture_type

Missing Required Inputs

FIELD	WHAT TO PROVIDE
product type	e.g. "air-source heat pump", "battery energy storage system"
target cost	Target unit manufacturing cost in GBP
production volume	Annual production quantity
max mass	Maximum allowable mass in kg
jurisdiction	Target market (e.g. "UK", "EU", "US")
thermal capacity kw	Heating/cooling capacity in kilowatts
cop target	Coefficient of Performance target at specified conditions
refrigerant type	e.g. "R290", "R410A", "R32"
acoustic target dba	Maximum noise level in dB(A)
architecture type	e.g. "monobloc", "refrigerant_split", "hydronic_split"

Next Actions

- Complete missing brief fields
- Lock architecture type
- Re-run pipeline

Biggest Blocker:
Required field missing: product_type

Contents

1. Brief & Requirements

1.1 Overview & Context

Subject / Use Case

Retrofitting existing fossil-fuel heating systems in lightweight commercial buildings to provide highly efficient, low-carbon space heating and domestic hot water.

Mission Statement

To accelerate the decarbonization of UK small commercial buildings by delivering an affordable, ultra-low GWP R290 hydronic split heat pump.

Target Customers

Small to medium office buildings, high-street retail spaces, local cafes, light industrial units, and small schools.

Why Now (Market Timing)

Stringent F-Gas phasedowns, tightening of UK Building Regulations restricting new gas boiler installations, rising ESG commitments, and maturation of R290 compressor technology.

Full Synthesis Report

The development of a compact commercial 30kW air-source heat pump (ASHP) utilizing R290 (propane) refrigerant presents a highly lucrative but technically rigorous opportunity within the UK market. Driven by strict national decarbonization targets, the mandate to phase out fossil fuel boilers, and the tightening of the F-Gas regulations, the commercial transition toward ultra-low Global Warming Potential (GWP) refrigerants like R290 is accelerating rapidly. The specified target of achieving a COP of 3.5 or higher at A2/W35 is an ambitious but structurally feasible technical requirement that positions this product at the premium tier of energy efficiency, ensuring future-proof compliance with tightening Building Regulations Part L requirements. The architectural choice of a 'hydronic split' system—where the hermetically sealed R290 refrigeration cycle remains entirely within the outdoor unit to completely mitigate indoor flammability risks, while an indoor unit manages water circulation, backup heating, and controls—is the optimal safety and performance trajectory. Securing a unit production cost of £4,500 at an annual volume of 500 units will require aggressive supply

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

chain optimization and strategic sourcing of R290-optimized inverter-driven scroll compressors and microchannel heat exchangers. Crucially, obtaining MCS (Microgeneration Certification Scheme) certification is non-negotiable for the strict UK landscape; without it, commercial installers cannot claim essential government incentives, such as the Boiler Upgrade Scheme, or satisfy green financing requirements. The target capacity of 30kW perfectly aligns with the heating loads of small to medium commercial premises such as bespoke high-street retail shops, regional office spaces, and boutique hospitality venues, effectively bridging the vast product gap between strictly residential units and massive multi-megawatt industrial chillers.

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

1.2 Market & Competitors

Competitor Landscape

Vaillant

Product	**aroTHERM plus (Cascaded)**
Pricing Strategy	**£8,000 - £10,000+ for standard hardware cascade required to hit 30kW.**
Technical Specs	**R290 monoblocs, cascading residential units to achieve up to 35kW, COP ~3.6 at A2/W35.**

STRENGTHS

Phenomenally established UK network, highly reliable, strong user interface.

WEAKNESSES

Relying on multiple cascaded residential units significantly increases installation complexity, footprint, and total system cost.

DIFFERENTIATION ANGLE

Providing a single, native 30kW unit severely undercuts cascade hardware costs and halves installation footprint/time.

Samsung

Product	**EHS Mono HT Quiet R290**
Pricing Strategy	**£6,500 - £8,000 to cascade enough units for a 30kW load.**
Technical Specs	**R290 monobloc delivering high temperatures, currently capping single unit capacities around 16-20kW.**

STRENGTHS

Class-leading acoustic insulation, strong design aesthetic, and robust high-temperature delivery.

WEAKNESSES

Lacks a standalone 30kW unit, forcing expensive and complex multi-unit installations for small commercial loads.

DIFFERENTIATION ANGLE

Targeting 30kW in one discrete chassis achieves superior economies of scale and dramatically simpler hydronic balancing.

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

Mitsubishi Electric

Product	**Ecodan CAHV Series**
Pricing Strategy	**£12,000 - £15,000 per unit.**
Technical Specs	**43kW capacity commercial units, mostly utilizing R454C or older R407C, heavy commercial framework.**

STRENGTHS

Incredible industrial-grade durability and complete market dominance in UK commercial HVAC.

WEAKNESSES

Too large, very expensive, and relies on transition/higher-GWP refrigerants rather than truly future-proof R290.

DIFFERENTIATION ANGLE

Massively undercutting the industrial price point (£4,500 target) tailored precisely for 'small' commercial, utilizing better ESG-compliant refrigerants.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

1.3 Engineering Constraints

Core Targets

Unit Cost Ceiling	**£4,500.00**
Max Mass	**-**
Target Process	
Target Material	
Tolerance Target	
Production Quantity	
Batch Size	**500**

Research Sources

TITLE	TYPE	YEAR	RELEVANCE
MCS MIS 3005-I Standard	Standard	2026	Dictates the certification requirements for heat pump installations and product eligibility in the UK.
BSRIA UK Heat Pump Market Report 2023	Market Report	2026	Provides realistic volume projections, pricing structures, and adoption rates for commercial ASHPs in the UK market.
EN 14825 & EN 14511 Testing Methodologies	Technical Standard	2026	Outlines the rigorous testing methodologies required to legally declare the target COP of >=3.5 at A2/W35.
DSEAR Guidelines for R290 Systems	Regulation Guide	2026	Crucial for navigating the explosive atmosphere and workplace safety requirements associated with propane refrigerant.
Wholesale Compressor and Hydronic Component Catalogs	Supplier Data	2026	Essential benchmarking data to validate and manage the £4,500 target Bill of Materials for a 500-unit/year volume.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

Note: Some regulatory assessments (charge-limit calculations, DSEAR classification) require completed sizing data. These will be populated once sizing is complete.

2.1 MCS MIS 3005-I

Microgeneration Certification Scheme

Status: not-started | Owner: Certification Engineer

Standard Summary

Governs the performance standards and certification required to access UK heating subsidies.

Applicability Assessment

> UK-wide product sales and installation.

Engineering Design Impact

Requires specific seasonal performance targets and standard-compliant technical documentation.

EVIDENCE REQUIRED

Accredited laboratory test reports confirming COP and acoustic data.

GAP ACTION

Book product testing slot at an accredited test lab such as BSRIA or BRE.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.2 DSEAR 2002

Dangerous Substances and Explosive Atmospheres Regulations

Status: not-started | Owner: Safety & Compliance Lead

Standard Summary

Workplace safety legislation addressing the risks of flammable substances like R290.

Applicability Assessment

> Handling, manufacturing, and installation of the equipment.

Engineering Design Impact

Mandates intrinsically safe / ATEX-compliant electrical components within the outdoor unit's potential leak zones.

EVIDENCE REQUIRED

Risk assessments, zone modeling reports, and ATEX component certificates.

GAP ACTION

Conduct ATEX explosive zone modeling for the outdoor chassis design.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.3 EN 378

Safety and Environmental Requirements for Refrigerating Systems

Status: not-started | Owner: Mechanical Design Engineer

Standard Summary

European standard defining safety limits for refrigerant charges and equipment locations.

Applicability Assessment

> Refrigerant loop and placement of the outdoor unit.

Engineering Design Impact

Will require rigorous charge calculations and dictates ventilation spacing away from building openings.

EVIDENCE REQUIRED

Charge limit calculations and system pressure test validation.

GAP ACTION

Determine minimum safe distances and design proper enclosure ventilation mechanisms.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.4 ENA G99

Grid Connection Requirements

Status: not-started | Owner: Electrical Engineer

Standard Summary

UK specific grid connection regulations for generating and high-draw equipment over 16A per phase.

Applicability Assessment

> Electrical and inverter design.

Engineering Design Impact

Requires robust power quality controls, strict harmonic limits, and active power management.

EVIDENCE REQUIRED

Harmonic emission test reports and G99 compliance forms.

GAP ACTION

Design and validate the inverter filtering circuits against G99 harmonic constraints.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.5 UKCA/CE LV-EMC

Low Voltage and EMC Directives

Status: not-started Owner: Compliance Manager

Standard Summary

Vital standards for essential electrical safety and electromagnetic compatibility.

Applicability Assessment

Entire system electronics.

Engineering Design Impact

Necessitates strict grounding, advanced cable shielding, and isolation rules suitable for commercial power connections.

EVIDENCE REQUIRED

LVD and EMC laboratory test reports and Declaration of Conformity.

GAP ACTION

Schedule third-party EMC testing when the first full integrated prototype is complete.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

3. Sizing

BLOCKED — Sizing not generated

Sizing requires: thermal capacity, ambient temperature, flow temperature, and envelope dimensions. These fields are missing from the brief.

4. System Modules

BLOCKED — Modules not generated

Module architecture requires completed sizing and spatial allocation.

5. Bill of Materials

BLOCKED — BOM not generated

Bill of materials requires completed module architecture.

6. Cost Waterfall

BLOCKED — Cost analysis requires completed sizing and bill of materials

Unit cost cannot be calculated without component selection. Complete sizing and BOM generation first.

7. Risks Register

BLOCKED — Risks not generated

Risk register requires completed system modules.

8. Supplier Shortlist

BLOCKED — Suppliers not matched

Supplier matching requires a completed bill of materials.

9. Audit Log

Pipeline execution log — generated by PDF Engine v2.1.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.

Pipeline Execution Summary

Project Identifier	**a_compact_commercial_airsource_heat_pump**
Research Result	**Generated**
System Modules	**0 Modules**
Bill of Materials Lines	**0 Parts**
Cost Analysis	**Pending**
Specialist Reviews	**0 Completed**
Supplier Shortlists	**0 Matched**

10. Source Attribution

This table documents the origin of data for each major section of the report, ensuring traceability of LLM outputs versus deterministic or user-provided data.

SECTION	SOURCE	PROVIDER / SYSTEM	DETAIL
Research	LLM	OpenRouter	Gemini 3.1 Pro via OpenRouter
Research	USER	user	Founder brief text
Regulatory	LLM	Unknown LLM	Gemini 3.1 Pro — standards extraction